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Development and operation of alternative oxygen electrode materials for hydrogen production by high temperature steam electrolysis

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ABSTRACT

High temperature steam electrolysis (HTSE) is one of the most promising ways for hydrogen mass production. To make this technology suitable from an economical point of view, each component of the system has to be optimized, from the balance of plant to the single solid oxide electrolysis cell. At this level, the optimization of the oxygen electrode is of particular interest since it contributes to a large extent to the cell polarization resistance. The present paper is focused on alternative oxygen electrode materials with improved performances compared to the usual ones mainly based on perovskite structure. Two nickelates, with compositions $La_2NiO_{4+\delta}$ and $Nd_2NiO_{4+\delta}$ are investigated and evaluated in HTSE operation at the button cell level. The performances of the $Ln_2NiO_{4+\delta}$ - containing cells (Ln = La, Nd) is improved compared to a cell containing the classical Sr-doped $LaMnO_3$ (LSM) perovskite oxygen electrode showing that nickelates are promising candidates for HTSE oxygen electrodes, especially for operation below 800 °C. Indeed, current densities determined at 1.3 V are 1.1 times larger for the $La_2NiO_{4+\delta}$ - containing cell and 1.6 times larger for the $Nd_2NiO_{4+\delta}$ one compared to the LSM - containing cell at 850 °C, whereas at 750 °C they are 1.8 and 4.4 times larger, respectively. Thanks to the use of a reference electrode, by coupling impedance spectroscopy and polarization measurements, the overpotential of each working electrode is deconvoluted from the complete cell voltage under HTSE operating conditions.
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1. Introduction

For economical and ecological reasons, hydrogen is considered as a promising energetic vector for future [1]. In order to become a recognized alternative fuel, one crucial question still remains to be solved: How to produce massive hydrogen quantity using clean processes with low or no CO_2 emissions? Water electrolysis is one of the most promising ways to achieve this target. The thermodynamic advantages of high temperature steam electrolysis (HTSE) in comparison to the conventional low temperature electrolysis processes have been known for long [2]. The HTSE basic principle is to split water molecules into hydrogen and oxygen using electricity and heat, provided by renewable energy sources [3] or nuclear power plants [4–6] for example. However, in order to produce by HTSE the amount of hydrogen necessary to replace even

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partially the fossil fuels, many dedicated power plants would be necessary. Furthermore, to reduce this required energy, the development of highly efficient systems is necessary. Each system component has to be optimized, from the balance of plant to the materials constituting the solid oxide electrolysis cell (SOEC) where the HTSE reaction occurs.

As described in the case of solid oxide fuel cells (SOFCs), at the cell level, electrode polarizations induce large voltage losses that should be reduced for increasing efficiency [7]. The oxygen electrode is of particular interest as it contributes to a large extent to the cell polarization resistance. The use of mixed ionic and electronic conducting (MIEC) oxygen electrodes (either composite or single material) is beneficial since they extend the reaction zones, thus reducing the activation polarization and inducing better electrode performances [7]. Commercially available cells contain perovskite-based oxygen electrodes, mainly made of Sr-doped $LaMnO_3$ (LSM), which is a pure electronic conducting material. This paper reports alternative oxygen electrode materials with improved performances compared to the regular perovskites. The $Ln_2NiO_{4+\delta}$ - type oxides (with the K_2NiF_4 structure) were selected because of their aptitude to accommodate a large oxygen overstoichiometry. This capability leads to a mixed ionic and electronic conductivity under oxidizing conditions [8,9]. Such characteristics are already well known for a use in SOFCs and start to be studied under HTSE operating conditions [10]. In addition, under anodic polarization, it was observed that the $La_2NiO_{4+\delta}$ oxide exhibits a lower polarization resistance than under cathodic polarization [11]. The purpose of the present study is to investigate the lanthanum nickelate ($La_2NiO_{4+\delta}$) and to compare its performances with those of the recently studied $Nd_2NiO_{4+\delta}$ [12] and of a reference LSM based electrode to confirm the interest of the $Ln_2NiO_{4+\delta}$ compounds as compared to the perovskite ones for the HTSE application.

In addition, thanks to electrochemical impedance spectroscopy (EIS) and polarization measurements performed in a specific setup equipped with a reference electrode, anodic and cathodic contributions were separated from the global cell voltage under HTSE operating conditions.

2. Experimental

$La_2NiO_{4+\delta}$ and $Nd_2NiO_{4+\delta}$ oxygen electrode materials were tested on small size single cells. Such cells were planar circular shaped with two symmetrical electrodes of the same geometric area: 3.14 cm^2 (20 mm diameter).

Powders were prepared by nitrate-citrate route as described by Courty et al. [13]. Stoichiometric amounts of lanthanide and nickel oxides were dissolved in diluted nitric acid. After addition of a large excess of citric acid, the solution was dehydrated and heated until self-combustion of the precipitate to obtain submicronic precursor particles [8]. The final annealing was performed at 950 °C for 8 h for $La_2NiO_{4+\delta}$ and at 1000 °C for 12 h for $Nd_2NiO_{4+\delta}$ to obtain single phases. The powder was then ball milled to obtain a median grain size ($d_{0.5}$) of about 0.8 μm. A terpineol-based slurry was prepared from each powder for screen printing. $La_2NiO_{4+\delta}$ and $Nd_2NiO_{4+\delta}$ were deposited on commercial (H.C. Starck Ceramics GmbH & Co. KG, 50 mm diameter) half electrolyte supported cells made of gadolinium doped ceria-Ni cermet (Ni-CGO) hydrogen electrode and 3 mol% tetragonal yttria stabilized zirconia (TZ3Y) electrolyte, 90 μm thick. After sintering at 1100 °C for 3 h in air [14], the thickness of the final porous oxygen electrodes was about 30 μm (Fig. 1).

A gold annular reference electrode (2 mm wide) was painted on the oxygen electrode side of the electrolyte of each nickelate - containing cell and then annealed at 750 °C for 30 min in air (Fig. 2). The distance between working and reference electrodes (more than 100 times the thickness of the electrolyte) was set to avoid as much as possible measurement artifacts due to the influence of the current lines through the electrolyte on the reference electrode [15]. Cells were placed in a ceramic housing and sealed using glass paste and gold rings, then placed in a specific setup equipped with a so-called 3rd chamber under controlled atmosphere for the reference electrode (Fig. 2). The pressure in each room was 1 atm.

Performances of the nickelate oxygen electrode containing cells were compared to a reference commercial cell provided by H.C. Starck Ceramics GmbH & Co. KG (ESC2), made of the same electrolyte (TZ3Y, 90 μm thick) and the same Ni-CGO hydrogen electrode but containing an optimized LSM based oxygen electrode.

During electrochemical measurements the inlet gas composition to the hydrogen electrode was composed of hydrogen, argon and water steam which was obtained by total evaporation of liquid water. All the results reported in this study were recorded with a $p(H_2O)/p(H_2)$ in between 0.9 and 1.9. It was experimentally proved that results remain the same over this whole range. This important hydrogen quantity was set to avoid any Ni oxidation, even at the open circuit voltage (OCV). The inlet gas of the positive electrode chamber was air. Current collectors were made of platinum grids (3600 meshes cm^{-2}) directly in contact with the electrodes, covered with gold grids (1024 meshes cm^{-2}). Measurements were performed at 750 °C, 800 °C and 850 °C, from the lowest to the highest temperature.

Current density - voltage curves (i-V curves) were recorded from the OCV to the highest polarization, then back to the OCV. The i-V curves were obtained by applying a dc current

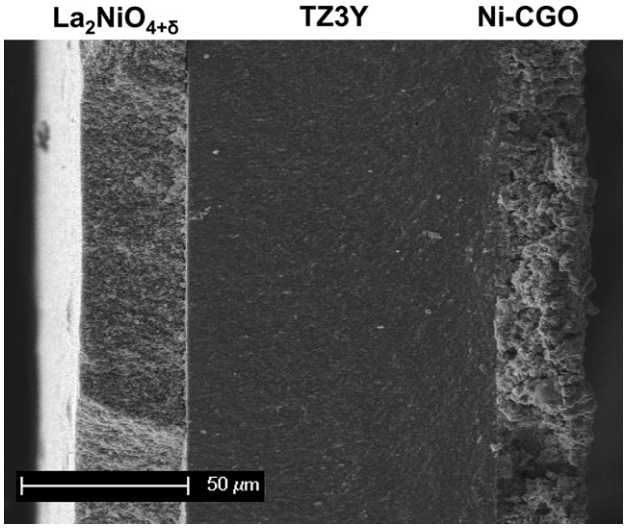


Fig. 1 – SEM micrograph of the $La_2NiO_{4+\delta}$/TZ3Y/Ni-CGO cell cross section after study.

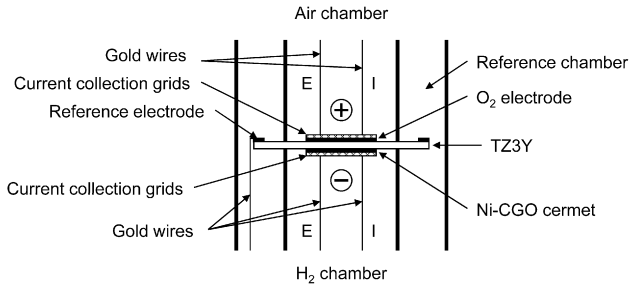


Fig. 2 – Schematic representation of the test rig.

between positive and negative electrodes during 180 s for each step and recording the voltage after stabilization. The performance of the cells was evaluated through the current density and the area specific resistances (ASR) at 1.3 V. ASRs were calculated as the slopes of i-V curves between 1.25 V and 1.35 V. The ratio of steam utilization is calculated as the molar amount of produced hydrogen (according to the Faraday's law and assuming 100% current efficiency) divided by the molar amount of water in the inlet hydrogen chamber gas.

The EIS measurements were also performed under galvanostatic control with a 100 mA modulation amplitude, from 10^4 to 10^{-1} Hz, using an Autolab PGSTAT 302N equipped with a frequency response analyzer (FRA).

3. Results and discussion

3.1. *Cell performances*

The i-V curves measured at 800 °C on nickelate - containing cells in comparison to the LSM - containing one are shown in Fig. 3. The current densities and ASR values measured on i-V curves at 1.3 V for the three cell types at 850, 800 and 750 °C are reported in Table 1. Data show that performances of cells

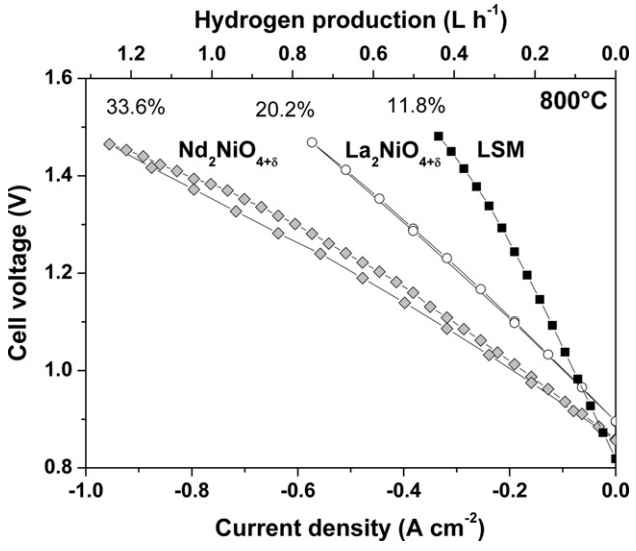


Fig. 3 – Current density - voltage curves measured at 800 °C on cells containing $La_2NiO_{4+\delta}$ (○), $Nd_2NiO_{4+\delta}$ (◇) or LSM (■); maximum steam utilization and hydrogen production are given in the graph.

Table 1 – Current densities and ASR values measured at 1.3 V for the three cell types at 850, 800 and 750 °C.

Oxygen electrode	Temperature (°C)	i at 1.3 V (A cm^{-2})	ASR at 1.3 V (Ω cm^2)
$La_2NiO_{4+\delta}$	850	−0.58	0.67
	800	−0.39	0.96
	750	−0.16	2.40
$Nd_2NiO_{4+\delta}$	850	−0.87[a]	0.41[a]
	800	−0.64[a]	0.57[a]
	750	−0.40[a]	0.93[a]
LSM	850	−0.53	0.83
	800	−0.22	1.87
	750	−0.09	4.24

a Average values of increasing and decreasing polarization measurements.

containing nickelates oxygen electrodes are improved compared to the LSM - containing one over the whole investigated temperature range.

The $Nd_2NiO_{4+\delta}$ - containing cell is the best performing one. Current densities of −0.87, −0.64 and −0.40 A cm^{-2} have been measured at 1.3 V for 850, 800 and 750 °C, respectively. The $La_2NiO_{4+\delta}$ - containing cell led to current densities of −0.58, −0.39 and −0.16 A cm^{-2} for the same three temperatures. Compared to LSM, current densities achieved at 850, 800 and 750 °C are 1.6, 2.9 and 4.4 times larger with $Nd_2NiO_{4+\delta}$ and 1.1, 1.8 and 1.8 times larger with $La_2NiO_{4+\delta}$. As a consequence, ASR values are much lower for the nickelate oxides - containing cells as reported in Table 1. Compared to the LSM - containing cell, at 850 °C they are 2.0 and 1.2 times lower for the $Nd_2NiO_{4+\delta}$ - containing one and for the $La_2NiO_{4+\delta}$ - containing one, respectively. At 800 °C, they are 3.3 and 1.9 times lower, and at 750 °C they are 4.6 and 1.8 times lower. These results do show that these nickelates are suitable as SOEC oxygen electrode material for the whole investigated temperature range, and particularly below 800 °C.

3.2. *Anode and cathode contributions*

The aim of this part is to separate the various contributions in one of the i-V curves shown in Fig. 3. The used methodology has recently been reported for the $Nd_2NiO_{4+\delta}$ - containing cell [12]. This section describes the same study for the $La_2NiO_{4+\delta}$ - containing cell.

The data presented in Fig. 3 were recorded by measuring the voltage between positive and negative electrodes. This total HTSE cell voltage is composed of different voltage losses as described by Eq. (1).

$$E_{cell} = E_{Nernst} - R_N \times i - \eta(Ni - CGO) + \eta(La_2NiO_{4+\delta}) \quad (1)$$

E_{Nernst} is the potential resulting from the difference in oxygen activity between the air and the hydrogen chambers (the activity being assimilated to the oxygen partial pressure), the $R_N \times i$ term is the voltage loss due to the ohmic drop (R_N being the resistance normalized by the geometric area of the electrodes). The two final terms represent the overpotentials η of each electrode. The i-V curves between anode and reference and between cathode and reference were also recorded.

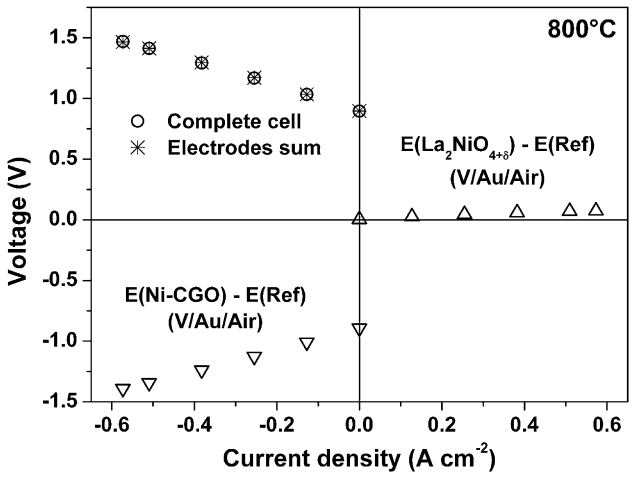


Fig. 4 – Current density - voltage curves recorded at 800 °C, between the oxygen electrode ($La_2NiO_{4+\delta}$) and the reference (△), between the hydrogen electrode (Ni-CGO) and the reference (▽), then compared with the measurements between the oxygen and hydrogen electrodes (○); stars (✱) correspond to the sum of the voltage of the first two contributions; reference electrode flowed with air.

They are plotted in Fig. 4 for 800 °C and compared to the i-V curve measured on the complete cell. The sum of $La_2NiO_{4+\delta}$/reference voltage (anodic contribution) and Ni-CGO/reference voltage (cathodic contribution) gives the same value as the one measured between the $La_2NiO_{4+\delta}$ and Ni-CGO electrodes (complete cell). It is clearly evidenced in Fig. 4, which confirms that the reference electrode is stable under polarization. The slope measured for the anodic i-V curve is smaller than the one measured for the cathodic one. Indeed, ASR values at 1.3 V are equal to 0.10 Ω cm^2 for the anodic curve and to 0.85 Ω cm^2 for the cathodic one.

As for the complete cell, the voltages measured with the reference electrode are composed of different voltage losses according to Eqs. (2) and (3).

$$E(La_2NiO_{4+\delta}) - E(Ref) = E^a_{Nernst} + R^a_N \times i^a + \eta(La_2NiO_{4+\delta}) \quad (2)$$

$$E(Ni - CGO) - E(Ref) = E^c_{Nernst} + R^c_N \times i^c + \eta(Ni - CGO) \quad (3)$$

In these relations, E^a_{Nernst} and E^c_{Nernst} correspond to the voltages due to the oxygen partial pressure difference between the different chambers (air/reference chambers and hydrogen/reference chambers respectively, *a* meaning anode and c meaning cathode). The sum of R^a_N and R^c_N is the complete cell normalized resistance R_N used in Eq. (1). By convention, in these relations and in Fig. 4, the current measured on the anodic side (i^a) is positive and negative on the cathodic one (i^c), with $i = i^c = -i^a$.

EIS measurements have been performed on the complete cell on one hand and on the anodic and cathodic sides on the other hand, using the reference electrode. Diagrams recorded at OCV for these three configurations at 800 °C are given in Fig. 5. As for the i-V curves, the sum of the diagrams recorded on anodic and cathodic sides (vector sum for each frequency) fit perfectly to the experimental diagram recorded between anode and cathode. The ohmic loss for the complete cell were then calculated by multiplying the current density applied to the cell by the series resistance measured by impedance spectroscopy on complete cell (in two electrodes configuration). For each current density, the electrodes overpotential was calculated by subtracting the ohmic losses deduced from impedance diagrams in three probes configuration from the i-V curves also measured in three electrodes configuration. The sum of the different contributions agrees pretty well with the experimental i-V curve measured on complete cell as reported in Fig. 6.

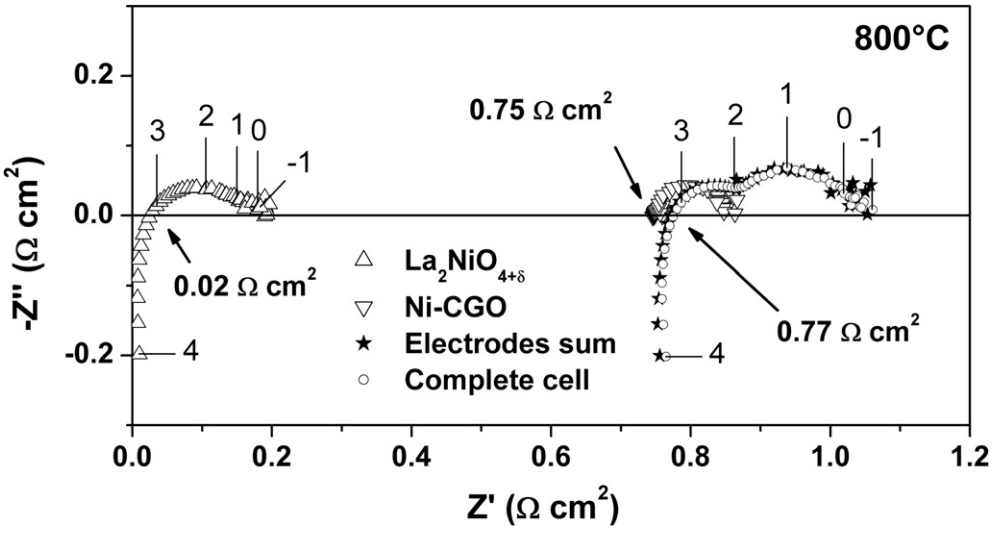


Fig. 5 – Impedance diagrams recorded at 800 °C at open circuit voltage between the oxygen electrode and the reference (△), between the hydrogen electrode and the reference (▽) and between the oxygen and hydrogen electrodes (○); reference electrode flowed with air. The stars (★) represent the vectorial sum of the two first contributions. The numbers by data points are frequency logarithms.

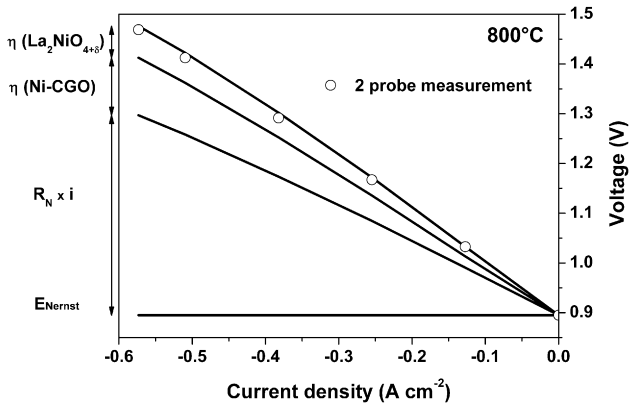


Fig. 6 – Current density - voltage curve measured at 800 °C for the $La_2NiO_{4+\delta}$ - containing cell (○) and split into ohmic loss ($R_N \times i$) and electrode overpotentials.

At 800 °C, a cell voltage (contributions sum) of 1.3 V corresponds to a current density of −0.38 A cm^{-2}. At this operating point, ohmic loss contribution is 275 mV (68%), η (Ni–CGO) is 80 mV (20%) and η ($La_2NiO_{4+\delta}$) is 50 mV (12%). These percentages remain almost constant over the whole investigated polarization range. As expected for an electrolyte supported cell, the ohmic drop contributes to the main part of the voltage loss. This electrolyte contribution can be easily reduced about 10 times by using electrode supported cells [16]. On the other hand, the $La_2NiO_{4+\delta}$ overpotential corresponds to the smallest measured contribution.

4. Conclusions

Two nickelate oxides with compositions $La_2NiO_{4+\delta}$ and $Nd_2NiO_{4+\delta}$ have been investigated as oxygen electrodes of SOEC and integrated in solid oxide electrolyte supported button cells made of a TZ3Y electrolyte and a Ni-CGO hydrogen electrode. The performances of these cells have been compared to a commercial electrolyte supported cell containing the same hydrogen electrode and electrolyte layers but with a conventional LSM based oxygen electrode.

Improved performances compared to this cell have been evidenced for both nickelates over the whole investigated temperature range (750–850 °C). At 850 °C, the current density at 1.3 V is 1.1 and 1.6 times larger with $La_2NiO_{4+\delta}$ and $Nd_2NiO_{4+\delta}$, respectively. At 800 °C, they are 1.8 and 2.9 times larger, and at 750 °C, they are 1.8 and 4.4 times larger. Such results point out that these materials are the most suitable, the lowest the temperature is, especially for the neodymium nickelate.

Contrary to what can be observed with LSM [17], it can be noticed that in HTSE operating conditions no delamination of the oxygen electrode has been evidenced when using $Nd_2NiO_{4+\delta}$ or $La_2NiO_{4+\delta}$.

Using a reference electrode, it was also possible to determine each cell voltage loss contribution under HTSE operation. It has been clearly found that the use of lanthanum (this work) or neodymium [12] nickelate oxides as oxygen electrode materials results in small overpotentials compared to all the other contributions in the complete cell voltage.

These features can be explained by the beneficial effect of using mixed conducting electrode materials, which allows the reaction zone to be delocalized over the entire electrode surface. One should also point out, as previously discussed in terms of structural aspects [9], that these materials are able to easily accept additional oxygen atoms while keeping their mixed ionic and electronic conducting properties. Conversely, perovskite materials are completely oxidized and behave as only electronic conducting electrodes. The fact that the neodymium compound exhibits best performances than the lanthanum one is not still explained.

In more general terms, these results obtained on single cells point out that the use of optimized cells integrating advanced new electrode materials is very important for increasing hydrogen production performances.

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